Smart Library Book Borrowing and Fine Tracking System for a Rwandan Community Library

Shema Brian (29728) - INSY 8311

1. Problem Statement

A community library in Rwanda uses paper cards to track book borrowing and returns. This causes three main problems. First, books are lost because there is no clear record of who has them. Second, late fines are calculated by hand, which causes mistakes and arguments with members. Third, there is no waiting list, so if a book is out, other members cannot know when it will be available.

2. Proposed Solution

I will build an Oracle database with PL/SQL that tracks borrowed books, calculates late fines automatically, and manages waiting lists. The database will block members who have 3 overdue books from borrowing again. It will automatically calculate the fine in RWF based on the number of days late. A dashboard will display the most borrowed books and unpaid fines.

3. Data Source

The main dataset is the Book-Crossing Dataset from Kaggle. It contains real book titles, authors, and publication years (over 278,000 books). Because the dataset does not include Rwandan library members, I will create realistic reference data for members, locations (Kigali, Huye), and fine rates in RWF. I will state this clearly in my documentation.

4. Planned Tables

MEMBER, BOOK, AUTHOR, BORROW_RECORD, FINE, WAITING_QUEUE, APP_USER, AUDIT_LOG. BORROW_RECORD links a MEMBER to a BOOK. FINE links to BORROW_RECORD. WAITING_QUEUE tracks members waiting for a specific BOOK.

5. Planned PL/SQL Features

- Procedure `borrow_book`: Saves a new borrowing record and checks if the member is allowed.

- Function `calculate_fine`: Calculates days late multiplied by 100 RWF.

- Trigger `trg_block_overdue`: Stops a member with 3 overdue books from borrowing.

- Trigger `trg_auto_fine`: Automatically creates a fine when a book is returned late.

- Trigger `trg_audit`: Writes every change into the AUDIT_LOG.

- Package `library_pkg`: Groups the main procedures and functions.

- Cursor: Monthly report of most borrowed books.

- DBMS_SCHEDULER job: Runs every night to check for newly overdue books.

6. Innovation

A Power BI dashboard connected to the Oracle database. It will show the most borrowed books, members with unpaid fines, and the current waiting list.

7. References

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